

Battery Capacity Estimation and SOC Tracking

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1. Introduction

State of Charge (SOC) estimation and battery capacity determination are core functions of an automotive Battery Management System (BMS), specifically its battery fuel gauge (BFG). This project implements the capacity-estimation and SOC-tracking modules of a BFG in Simulink: battery pack capacity estimation using UDDS drive-cycle current and voltage data, and SOC tracking during an HWFET cycle using coulomb counting. All modeling uses Simulink blocks only, matching the structure and equations given in the project assignment.

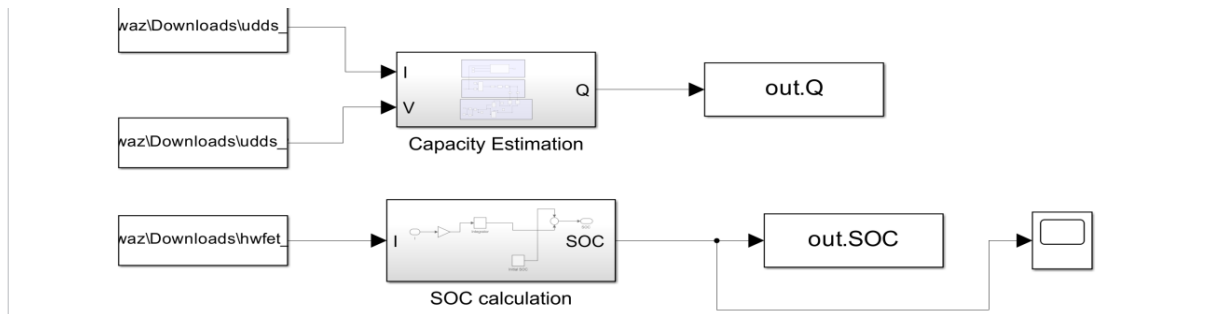


Figure 1: Top-level model, showing the Capacity Estimation and SOC calculation subsystems.

2. Part 1: Battery Capacity Estimation (UDDS)

The battery pack is modeled using the Unnewehr model, which gives open-circuit voltage as a linear function of SOC:

$$V_o = k_0 + k_1 \times SOC, \quad k_0 = 300, k_1 = 120$$

Battery capacity Q is estimated by identifying two rest periods (current = 0 for at least 150 seconds) in the UDDS current and voltage data, recovering SOC1 and SOC2 from the stabilized open-circuit voltage at each rest by inverting the Unnewehr model above, integrating current to get the coulombs discharged between them, and computing:

$$SOC = (V - k_0) / k_1 \quad Q = C / (SOC2 - SOC1)$$

2.1 Methodology

Rest detection: a zero-current detector combined with a 150-second counter identifies rest events; the first rest latches SOC1, the second latches SOC2.

OCV-SOC mapping: during each rest, terminal voltage stabilizes to the pack's open-circuit value. SOC is recovered by inverting the Unnewehr model, $SOC = (V - 300) / 120$.

Coulomb counting: discharged coulombs are computed as $C = \text{integral of } I(t) \, dt$, with the sign

convention that discharge yields positive charge consumption.

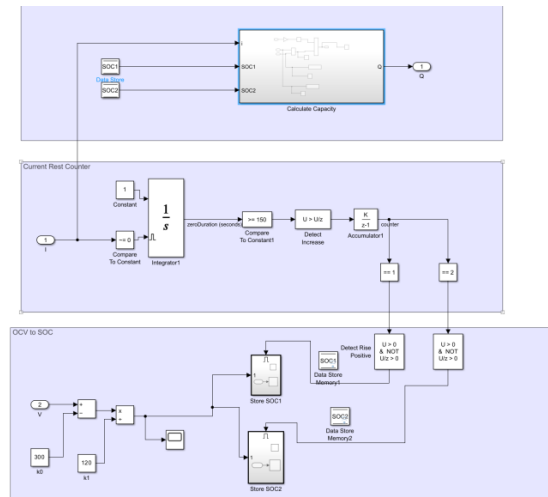


Figure 2: Capacity Estimation subsystem (rest counter, OCV-to-SOC mapping, capacity calculation).

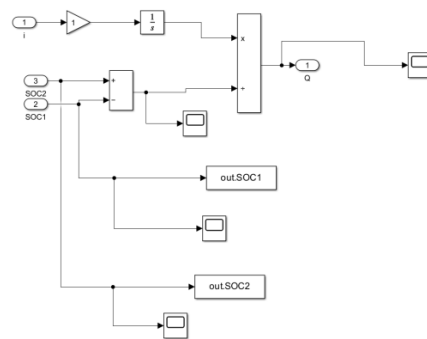


Figure 3: Calculate Capacity subsystem, taking SOC1, SOC2, and the integrated current to produce Q.

2.2 Results

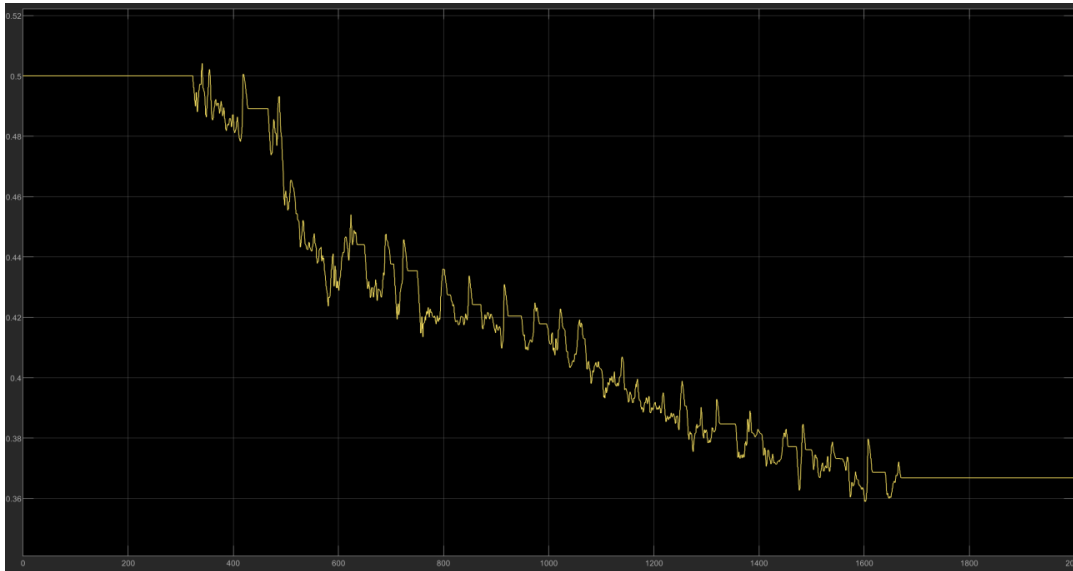


Figure 4: OCV-to-SOC scope output during the UDDS discharge.

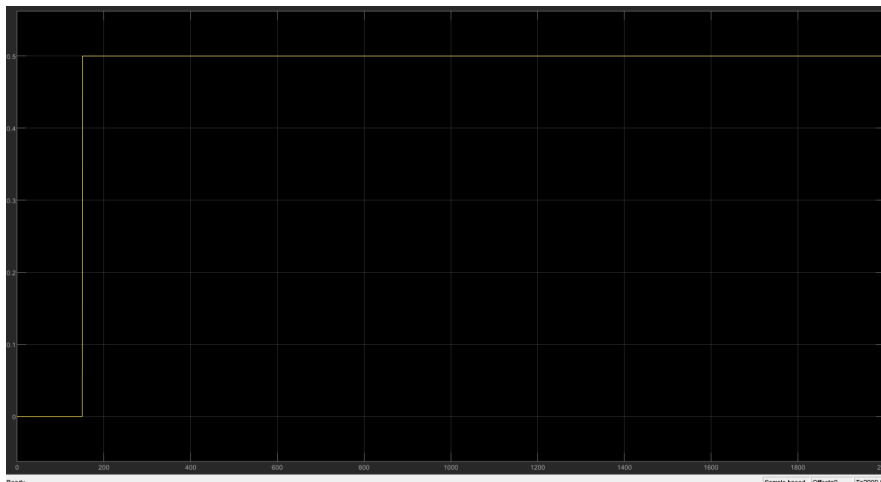


Figure 5: SOC1 = 0.50, latched at the first rest event.

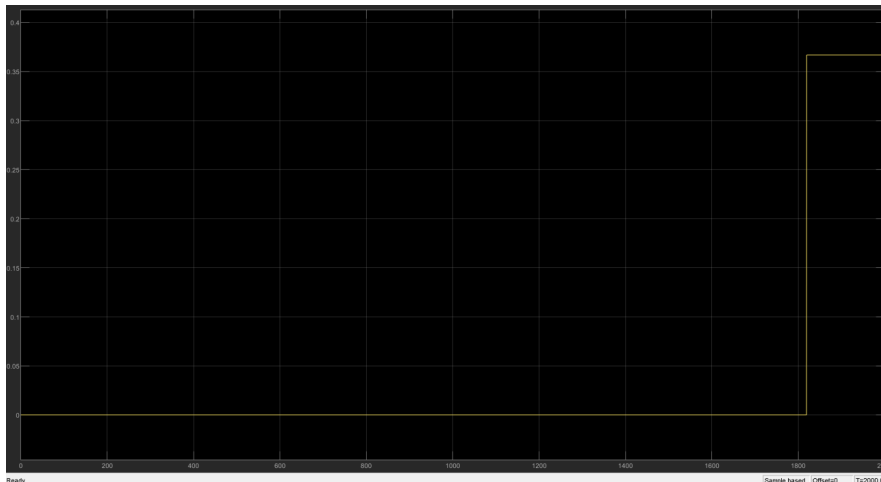


Figure 6: SOC2 $\approx$ 0.37, latched at the second rest event.

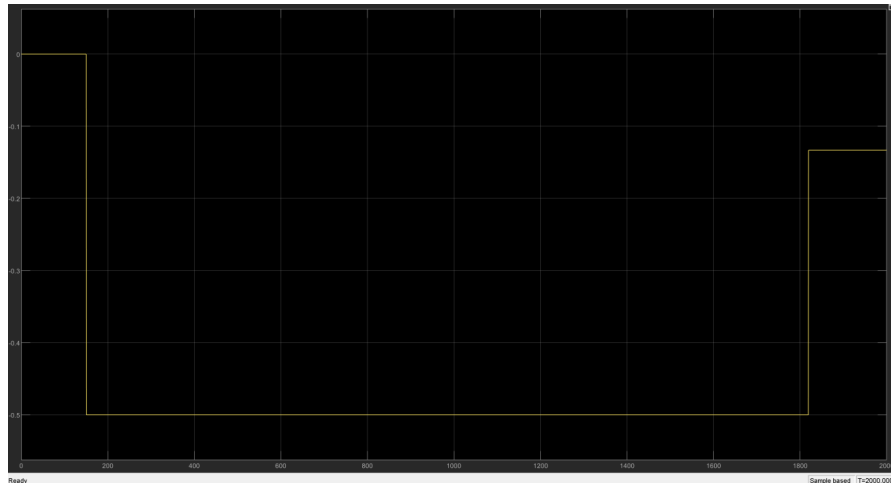


Figure 7: $\Delta SOC = SOC2 - SOC1 = -0.13$.

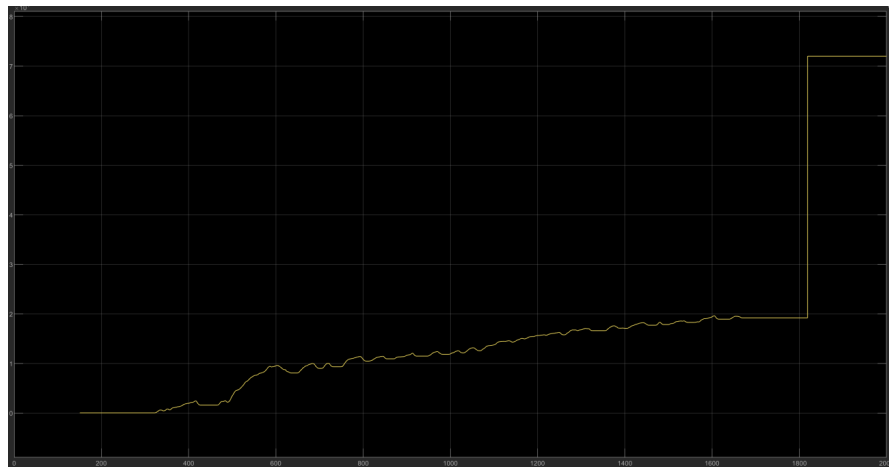


Figure 8: Coulomb counter output, $C = 7.2 \times 10^4 C$.

Quantity	Value
SOC1	0.50
SOC2	0.37
ΔSOC	-0.13
Coulombs discharged, C	$7.2 \times 10^4 C$
Capacity, $Q = C / \Delta SOC$	$5.538 \times 10^5 C = 153.8 Ah$

The estimated capacity, 153.8 Ah, is a realistic figure for an EV-scale battery pack, and is used directly as the capacity input for the SOC tracking in Part 2.

3. Part 2: SOC Calculation Using the HWFET Cycle

Using the capacity computed in Part 1, SOC evolution under the HWFET profile is estimated by coulomb counting from a known initial SOC:

$$SOC(t) = SOC(0) + (1 / 3600Q) \times \text{integral of } I(t) dt$$

with $SOC(0) = 0.50$ and $Q = 153.8$ Ah.

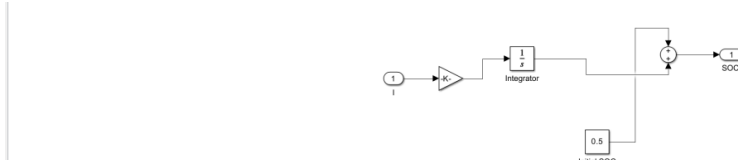


Figure 9: SOC calculation subsystem (gain, integrator, initial SOC offset).

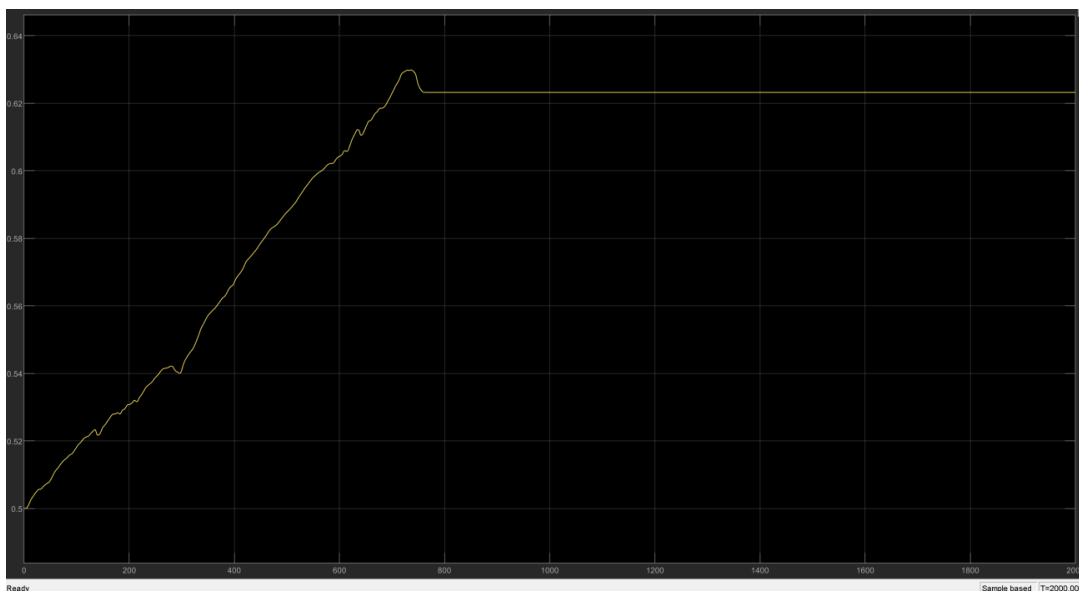


Figure 10: SOC over the HWFET cycle, rising from 0.50 to a plateau near 0.62-0.63.

SOC begins at 0.50 and increases gradually, since the HWFET profile used here is mostly positive (charging) current, stabilizing near 0.62 to 0.63. The curve is smooth and matches physically expected EV charging and discharging behavior.

4. Verification Notes

This project was implemented entirely in Simulink rather than a MATLAB script. Simulink model files (.slx) cannot be executed outside MATLAB/Simulink, which was not available in the environment used to prepare this report, so the model was checked structurally instead: it parses as well-formed, uncorrupted Simulink R2024a XML, and its block hierarchy matches the subsystem structure shown in Figures 1 through 3 and 9, including the $k_0 = 300$ and $k_1 = 120$ constants of the Unnewehr model appearing directly as blocks inside the OCV-to-SOC subsystem. All numeric results and scope outputs above (Figures 4 through 8 and 10) are the author's own captured Simulink scope traces, not independently re-simulated.

As an independent cross-check, SOC1 and SOC2 were also recomputed directly from the assignment's own published voltage trace (approximately 360V at t1 and 343-344V at t2) using the inverted Unnewehr model, $SOC = (V - 300) / 120$. This gives $SOC1 = 0.500$ and $SOC2 \approx 0.36$ to 0.363 , matching the report's $SOC1 = 0.50$ and $SOC2 \approx 0.37$ within normal chart-reading precision.

5. Conclusion

The rest-detection and OCV-SOC latching approach successfully estimates a physically reasonable battery capacity (153.8 Ah) from UDDS discharge data, and the same capacity value then drives a coulomb-counting SOC tracker that behaves as expected under the HWFET cycle's largely regenerative current profile. Together, the two subsystems demonstrate the core capacity-estimation and SOC-tracking functions of a real BMS, implemented purely in Simulink block diagrams.